

Marks Economy Architecture, Regulatory Compliance & Engineering Blueprint

Prepared for Cogito Academy Digital Platform Development & Compliance | August 2026

1. Marks Economy & Package Architecture

The Cogito digital platform implements a closed-loop internal currency system ("Marks") designed to streamline class bookings, gamify student engagement, and optimize unit economics while strictly adhering to Indonesian financial regulatory frameworks.

1.1 Core Conversion & Structural Mechanism

- **Platform Base Calculation Value:** 1 Mark = IDR 5,000. This base rate serves as the standard computational unit for class pricing, platform take rates, and conversion math across all interfaces.
- **Volume-Tiered Acquisition Pricing:** Students purchase Marks packages with rates ranging from **IDR 6,250 / Mark** down to **IDR 5,000 / Mark**.
- **Platform Breakage & Spread Margin:** Because all class costs and platform fees are calculated at 1 Mark = IDR 5,000, any premium paid upfront on smaller tiers (e.g., IDR 6,250 or IDR 5,750 per Mark) represents immediate, non-dilutive platform spread retained fully by Cogito Academy upon purchase.
- **Closed-Loop Utility Character:** Marks represent prepaid learning credits / digital vouchers. Once purchased, they can only be exhausted internally to book classes or satisfy access thresholds (e.g., Knowledge Bank). Marks hold zero cash surrender value and are strictly non-refundable and non-transferable.

1.2 Revised Student Purchase Packages

Package Name	Amount of Marks	Price per Mark (IDR)	Total Price (IDR)	Platform Spread Margin
Starter Pack	50 Marks	IDR 6,250	IDR 312,500	IDR 62,500 (20.0%)
Learner Pack	120 Marks	IDR 5,750	IDR 690,000	IDR 90,000 (13.0%)

Package Name	Amount of Marks	Price per Mark (IDR)	Total Price (IDR)	Platform Spread Margin
Explorer Pack	200 Marks	IDR 5,350	IDR 1,070,000	IDR 70,000 (6.5%)
Pioneer Pack	400 Marks	IDR 5,000	IDR 2,000,000	IDR 0 (Volume Baseline)

2. Take Rates, Tutor Compensation & Scenario Analysis

2.1 Mathematical Formulas & Constraints

- **Group Capacity Constraint:** Class size is strictly capped at $N \in \{1, 2, 3, 4, 5, 6\}$.
- **Tutor Honorarium Formula (T_{IDR}):**
 - *Online Class:* $T_{\text{Online}}(N) = \text{Base Rate} + (N - 1) \times \text{IDR } 30,000$ (Default 1-student baseline: IDR 175,000).
 - *Offline Class:* $T_{\text{Offline}}(N) = \text{Base Rate} + (N - 1) \times \text{IDR } 40,000$ (Default 1-student baseline: IDR 225,000).
 - *Step Interval Constraint:* Tutors may adjust their 1-student base rate only in increments of **IDR 5,000**.
- **Cogito Platform Take Rate Formula (C_{IDR}):**
 - *Online Class:* $C_{\text{Online}}(N) = \text{IDR } 50,000 + (N - 1) \times \text{IDR } 20,000$.
 - *Offline Class (Campus):* $C_{\text{Offline}}(N) = \text{IDR } 90,000 + (N - 1) \times \text{IDR } 40,000$.
- **Class Total & Per-Student Calculation:**
 - Total Pooled IDR: $\text{Total IDR}(N) = T_{\text{IDR}}(N) + C_{\text{IDR}}(N)$
 - Total Marks Pooled: $\text{Total Marks} = \frac{\text{Total IDR}(N)}{5,000}$
 - Marks Charged per Student: $\text{Marks per Student} = \left\lceil \frac{\text{Total Marks}}{N} \right\rceil$

2.2 Baseline Class Modeling

A. Standard Online Class (Tutor Baseline: IDR 175,000 | +IDR 30k/student)

Size (\$N\$)	Tutor Honorarium	Cogito Fee	Total IDR	Total Marks	Marks / Student	Actual Marks Pooled
1	IDR 175,000	IDR 50,000	IDR 225,000	45	45 Marks	45 Marks
2	IDR 205,000	IDR 70,000	IDR 275,000	55	28 Marks	56 Marks
3	IDR 235,000	IDR 90,000	IDR 325,000	65	22 Marks	66 Marks
4	IDR 265,000	IDR 110,000	IDR 375,000	75	19 Marks	76 Marks
5	IDR 295,000	IDR 130,000	IDR 425,000	85	17 Marks	85 Marks
6	IDR 325,000	IDR 150,000	IDR 475,000	95	16 Marks	96 Marks

B. Standard Offline Class at Cogito Campus (Tutor Baseline: IDR 225,000 | +IDR 40k/student)

Size (\$N\$)	Tutor Honorarium	Cogito Fee	Total IDR	Total Marks	Marks / Student	Actual Marks Pooled
1	IDR 225,000	IDR 90,000	IDR 315,000	63	63 Marks	63 Marks
2	IDR 265,000	IDR 130,000	IDR 395,000	79	40 Marks	80 Marks
3	IDR	IDR	IDR	95	32 Marks	96 Marks

Size (\$N\$)	Tutor Honorarium	Cogito Fee	Total IDR	Total Marks	Marks / Student	Actual Marks Pooled
	305,000	170,000	475,000			
4	IDR 345,000	IDR 210,000	IDR 555,000	111	28 Marks	112 Marks
5	IDR 385,000	IDR 250,000	IDR 635,000	127	26 Marks	130 Marks
6	IDR 425,000	IDR 290,000	IDR 715,000	143	24 Marks	144 Marks

2.3 Extreme Price Point Scenarios

- **Extreme Low End (Junior Coach Online @ IDR 75,000 Base):**
 - *1-Student:* Tutor IDR 75,000 + Cogito IDR 50,000 = IDR 125,000 (**25 Marks**).
 - *4-Student:* Tutor IDR 165,000 + Cogito IDR 110,000 = IDR 275,000 (**55 Marks total / 14 Marks per student**).
 - *Platform Safety:* Cogito's base operational margin remains completely protected regardless of how low a tutor prices their service.
- **Extreme High End (Masterclass / Specialist Offline @ IDR 500,000 Base):**
 - *1-Student:* Tutor IDR 500,000 + Cogito IDR 90,000 = IDR 590,000 (**118 Marks**).
 - *6-Student:* Tutor IDR 700,000 + Cogito IDR 290,000 = IDR 990,000 (**198 Marks total / 33 Marks per student**).
 - *Value Delivery:* Despite high tutor rates, per-student pricing drops down to 33 Marks (~IDR 165,000) for a premium offline workshop.

3. Regulatory & Legal Risk Analysis (Bank Indonesia Compliance)

3.1 Previous Regulatory Exposure

Under Bank Indonesia regulations (PBI No. 22/23/PBI/2020 on Payment Systems and PBI No. 23/6/PBI/2021 on Payment Service Providers / PJP), any system where users top up internal balances that are subsequently cashed out / withdrawn by third-party instructors constitutes *Penatausahaan Sumber Dana* (Administration of Sources of Funds / E-Money). Operating such a model requires a Category 1 PJP license with severe requirements (e.g., minimum paid-up capital of IDR 15 Billion and rigorous compliance certifications).

3.2 Legal Structure of the Closed-Loop Implementation

1. **Unilateral Closed-Loop Prepaid Utility:** Marks are legally defined in the platform Terms of Service as *non-refundable digital access credits / study vouchers*. Marks have zero cash surrender value and cannot be transferred P2P or withdrawn back to fiat currency by students.
2. **Severance of Direct Inter-User Monetary Exchange:** Tutors do not receive, hold, or convert Marks. Tutors operate under an independent service/honorarium agreement paid exclusively in IDR directly from Cogito's corporate operating account via standard banking/disbursement APIs.
3. **Access Gating Legitimacy:** Requiring a minimum balance (e.g., 35 Marks) to access the Knowledge Bank functions legally as an unencumbered *loyalty/membership tier rule*, since the student retains full ownership and spending power over those Marks at all times.
4. **Revenue Accounting:** Student top-ups are recognized as upfront deferred revenue for digital services, while tutor payouts are booked as cost of goods sold (COGS / instructor fees).

4. Actionable Engineering Roadmap: Mechanistic Changes

To implement this regulatory and economic transition seamlessly, the engineering team must execute the following structural and UI/UX adjustments:

4.1 Tutor Interface & Database Refactoring

- **Remove Marks Conversion from Tutor Views:** Eliminate any reference to Marks, "IDR 7,000 conversion rate", or coin wallets on the tutor dashboard. Tutors must interact exclusively in **IDR (Rupiah)**.
- **Rupiah-Denominated Base Input:** Replace the old Marks-based pricing form with an IDR input field constrained to steps of **IDR 5,000** (e.g., `<input type="number" step="5000" min="50000">`).
- **Dynamic Honorarium Breakdown Table:** The tutor pricing page should automatically render their exact earnings across all class sizes (1 to 6 students) using the incremental formula:

```
// Online Earnings Calculator  
TutorEarnings_Online(N) = BaseRate_IDR + ((N - 1) * 30000);  
  
// Offline Earnings Calculator  
TutorEarnings_Offline(N) = BaseRate_IDR + ((N - 1) * 40000);
```
- **Payout Module Renaming:** Change tutor withdrawal terminology in UI and code from "Withdraw Marks / Cash Out Coins" to **"Claim Honorarium / Payout Request"**, reflecting a standard B2B contractor payout.

4.2 Student Booking Engine & Frontend Refactoring

- **Automated Fee Aggregation:** When a student selects a class size \$N\$ and tutor, the backend must calculate:

```
const cogitoTakeRate = isOffline  
  ? 90000 + ((N - 1) * 40000)  
  : 50000 + ((N - 1) * 20000);
```

```
const totalIDR = tutorBaseRate + tutorIncrement + cogitoTakeRate;
const totalMarks = Math.ceil(totalIDR / 5000);
const marksPerStudent = Math.ceil(totalMarks / N);
```

- **Transparent Single-Metric Display:** Students only see the final **Marks per Student** required to join the session, alongside clear indicators of class size ($N = 1 \dots 6$).
- **Knowledge Bank Gating Logic:** Maintain the check `student.wallet.marks >= 35` for access, and ensure the UI explicitly states: *"Knowledge Bank is unlocked while your account holds at least 35 Marks (no Marks are deducted to view)."*

4.3 System Architecture Summary

Domain	Old Implementation	New Mandated Implementation
Tutor Dashboard	Tutor sets price in Marks (with IDR 7,000 cash-out).	Tutor sets price in IDR (increments of IDR 5k). Receives honorarium in IDR.
Student Dashboard	Purchased Marks at variable high tiers (8.5k-7.25k).	Purchases closed-loop Marks (IDR 6.25k - 5.0k). Consumes Marks on booking.
Financial Flow	Peer-to-peer coin transfer (Triggers BI e-money license).	Closed-loop voucher redemption + Direct contractor disbursement.